THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE CHAPTER TEST SET A

Chapter 13: **Our Home Earth, a Unique Life-Sustaining Planet** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Earth supports life mainly because of its: 1

(a) colour from space
(b) size of oceans only
(c) number of moons
(d) distance from the Sun, atmosphere and water

2. If Earth's gravity were too weak, the atmosphere would: 1

(a) turn to water **(b)** escape into space
(c) become denser **(d)** form ozone

3. The magnetic field of Earth shields life from: 1

(a) earthquakes **(b)** tides
(c) rainfall **(d)** high-energy particles from space

4. The hydrosphere consists of: 1

(a) rocks **(b)** living things **(c)** water **(d)** air

5. Reproduction is essential for: 1

(a) continuity of life **(b)** making oxygen
(c) forming ozone **(d)** changing seasons

6. The zygote is: 1

(a) an egg
(b) a seed coat
(c) a sperm
(d) the first cell formed after fertilisation

7. Which animal can regrow its body from a small piece? 1

(a) Bird **(b)** Elephant **(c)** Planaria **(d)** Cow

8. In most mammals, the embryo develops: 1

(a) in water **(b)** inside the female's body
(c) in soil **(d)** outside in eggs

9. Greenhouse gases mentioned include carbon dioxide and: 1

(a) helium **(b)** neon **(c)** methane **(d)** argon

10. Biodiversity is important because it: 1

(a) stops reproduction **(b)** makes the Earth smaller
(c) makes ecosystems stronger **(d)** removes microbes

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. What is the habitable zone? 2

12. Name two features of Earth that make it suitable for life. 2

13. Why is Earth's gravity important for life? 2

14. What is the function of the ozone layer? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 30–40 words each.

15. Explain why Earth's distance from the Sun is important. 3

16. Describe three features of Earth that make it habitable. 3

17. Explain the greenhouse effect and why it is important. 3

OR

Compare Venus and Earth in terms of greenhouse effect. 3

18. How do the four spheres interact? Explain with an example. 3

OR

Why is soil important for life? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 80–100 words each.

19. Explain the reasons why Earth is a unique planet that supports life. 5

OR

Describe the greenhouse effect, its natural role and how human activities enhance it. 5

20. Explain the four spheres of Earth and their interactions. 5

OR

Describe asexual reproduction in plants and animals with three examples. 5

END OF QUESTION PAPER Total: 40 Marks

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